Ayoub Mohamed

Fairfax County, VA | (703) 231-7893 | ayoubsmohamed07@gmail.com
github.com/powplowdevs | linkedin.com/in/ayoub-mohamed-3179112a2 | powplowdevs.github.io

EDUCATION

George Mason University, Fairfax, VA — Expected May 2029
B.S. in Information Technology, planned concentration in Software Solutions | GPA: 3.40
Relevant Coursework: Database Fundamentals, IT Architecture Fundamentals

SKILLS

Languages: C++, Go (Golang), Python, Java, JavaScript, SQL, HTML/CSS, object-oriented programming | Familiar: C, C#
Security & Networking: TLS/mTLS, X.509 certificates and PKI, TCP/IP sockets, asynchronous I/O, authentication and access control, bcrypt password hashing, rate limiting, audit logging
Libraries & APIs: Boost.Asio, Win32 API, WebExtensions API, zlib
Tools & Platforms: Git, Linux/Unix, MySQL, Raspberry Pi, Windows, Unity, VS Code

PROJECTS

ASIO Client-Server Framework | *C++, Boost.Asio, TCP* | GitHub — 2025 - Present
- Built a header-only C++ networking library on Boost.Asio for asynchronous TCP messaging between one server and multiple clients, using async accept, read and write calls
- Framed each message with a 4-byte length prefix so the receiver always reads complete messages, and queued incoming messages for the application to pick up

Zero-Trust mTLS Server | *Go, mTLS, X.509* | GitHub — 2026
- Built a zero-trust HTTPS server in Go that requires a client certificate signed by its own CA on every connection and ties each session to that certificate's SHA-256 fingerprint, so a stolen cookie fails on other devices
- Designed CSR-based device enrollment so private keys never leave the user's device, with admin approval before the CA signs anything, 5 assignable permissions, and device revocation
- Protected logins with certificate-plus-password checks, bcrypt hashing, per-IP rate limiting and 32-byte random session IDs in HttpOnly, Secure, SameSite=Strict cookies, and logged every login and admin action to a JSON-lines audit log

AutoSFX Wizard | *Python, C++, Win32 API, zlib* | GitHub — 2024 - 2025
- Built a tool that bundles files into one self-extracting Windows .exe: a Python CLI appends them to a prebuilt C++ stub with a JSON file table holding each file's compression, run order, run count and hidden settings
- Wrote the C++ stub that reads a 12-byte trailer from the end of its own .exe to find the file table, then extracts each file to a temp folder, streaming zlib-compressed ones through inflate in 16 KB chunks

SpyView | *JavaScript, WebExtensions API* | GitHub — 2025
- Published a Firefox extension that inspects every outgoing request through the webRequest API and flags requests to domains in DuckDuckGo's Tracker Radar, rating each tracker's risk from 1 to 5
- Built a popup that groups flagged trackers by the company that owns them, and a stats page with totals, top trackers and sites, recent history, a world map of tracker server locations and JSON export

One Life One Goal | *Unity, C#* — 2020
- Designed and released a 3D first-person shooter with 40 campaign levels, a survival mode, enemy AI and an in-game store, reaching 1,000+ downloads on itch.io and building a 2,000-follower community on X

EXPERIENCE

Peer Programming Tutor (Volunteer) — Sep 2022 - Jun 2025
Computer Science Honor Society, Carter G. Woodson High School, Fairfax, VA
- Tutored AP Computer Science A (Java) students one-on-one and in groups for 60 volunteer hours over three years, using guided problem-solving and code walkthroughs to explain recursion, data structures and debugging

CERTIFICATIONS

Foundations of Cybersecurity; Play It Safe: Manage Security Risks (Google, Coursera) — 2023